HEO KUAN SHUN

Senior Semiconductor Failure Analysis Engineer | Electrical FA, ATE Diagnostics & Yield Improvement

Malaysian / Singapore PR · Open to Relocation

+65 8518 9528 (SG) | ksheo9292@gmail.com | linkedin.com/in/ks9292 | ksheo9292.github.io/HKSCV2

PROFESSIONAL SUMMARY

Senior Semiconductor Failure Analysis Engineer with 10 years of experience in electrical fault isolation, physical failure analysis, product engineering, and high-volume semiconductor manufacturing. Hands-on in ATE debugging using Advantest V93K and Teradyne UFLEX; scan and memory diagnostics, and electrical-to-physical correlation. Proven in root-cause identification, reliability improvement, and cross-functional case closure with Product, Design, RMA, Quality, and Operations teams. Building Python, data analytics, and machine-learning capabilities for yield and defect-analysis applications.

CORE TECHNICAL SKILLS

Electrical FA & ATE (Semiconductor Silicon level)	Advantest V93K, Teradyne UFLEX, scan diagnostics, memory diagnostics, hotspot analysis, curve tracer, PEM, HRI, IR thermal / lock-in, laser-based localization, micro-probing
Physical FA	SEM, FIB, X-ray, CSAM, EDX, FTIR, EBSD, cross-section, delayering, decapsulation, material and structural defect analysis
Product / Yield Engineering	Wafer-sort yield analysis, DOE, FMEA, NPI, yield excursion investigation, root-cause analysis, corrective actions, customer returns, RMA
Data & Automation	Python, Pandas, SQL, data visualization, TensorFlow

EXPERIENCE

Electrical Engineer | Toshiba | Singapore — Aug 2026 - Present

- Performed PCB and circuit-level testing and troubleshooting to investigate electrical issues.
- Used oscilloscopes to measure supply voltages, power-supply ripple, and electrical signals during testing and fault investigation.
- Supported circuit design, modifications, and component evaluation through electrical testing and verification.
- Prepared test reports, verification records, and technical documentation to capture findings and support engineering reviews.

Professional Development | Self-Directed | Singapore — Apr 2025 - Sep 2026

- Studied AI, Machine Learning and Data Science through NUS Advanced Computing for Executives; final completion confirmation in Sept 2026, pending digital certification.

Senior Failure Analysis Engineer | Qualcomm - Mobile FA Team | Singapore — 2021 - 2025

- Led end-to-end electrical and physical failure analysis for high-volume RF and connectivity products, resolving approximately 80 cases annually with approximately 99% successful root-cause identification.
- Performed ATE-based electrical fault isolation using Advantest V93K and Teradyne UFLEX, applying scan diagnostics, memory diagnostics, hotspot analysis, and failure-signature correlation.
- Localized die-level failure regions using Photon Emission Microscopy (PEM) and high-resolution infrared thermal analysis, then correlated findings with physical FA results.
- Validated root causes through X-ray, CSAM, cross-sectioning, decapsulation, probing, and targeted physical inspection.
- Improved EFI sample-preparation and mounting processes, increasing repeatability and reducing analysis turnaround time.
- Partnered with Product, Design, RMA, and FA Operations teams to drive root-cause closure, corrective actions, technical reporting, and knowledge sharing.

Failure Analysis Engineer | Quasi-S Pte. Ltd. | Singapore — 2018 - 2021

- Performed advanced physical failure analysis using SEM, FIB, CSAM, EDX, X-ray, cross-sectioning, delayering, and decapsulation to identify structural and material defects.
- Executed device-level fault-isolation workflows for customer-return and reliability cases, supporting root-cause investigations and corrective-action decisions.

Product Engineer | Ardentec Singapore Pte. Ltd. | Singapore — 2016 - 2018

- Established standard operating flows and work instructions for wafer-sort manufacturing; conducted FMEA to improve process robustness and manufacturing readiness.
- Reviewed wafer-test yield trends, investigated yield-impacting issues, and collaborated with engineering teams to maintain and improve product yield.
- Developed and executed DOE plans for device-testing issues; supported New Product Introduction (NPI), including device startup, qualification, and production transfer.

- Used ATE platforms including Magnum, Teradyne J750, and Advantest V93K for test-related product engineering activities.

Technical Intern | Intel | Malaysia **2015**

- Validated and improved diagnostic solutions for customer diagnosis tools; explored IC capabilities, Arduino sketch-code development, and prototype circuit design.

EDUCATION & CERTIFICATIONS

- AI, ML & Data Science Programme | NUS School of Computing, Advanced Computing for Executives | Final completion confirmation in Sept 2026, pending digital certification.

- B.Eng. Electrical & Electronic Engineering | Universiti Malaysia Sabah | CGPA 3.55 / 4.00 | 2016